

Mohab Mohamed Rabie Mohamed

Artificial Intelligence Engineer | R&D Trainee Candidate

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<https://www.youtube.com/@HobaST/playlists>

PROFESSIONAL SUMMARY

Dynamic and results-driven Artificial Intelligence Engineer and 2025 Computer and AI graduate with an Excellent with Honors degree from Helwan University, currently pursuing a Master's in AI at Cairo University. Specialized in designing enterprise-grade RAG systems, fine-tuning and adapting pre-trained large language models (LLMs like GPT, LLAMA), and developing high-performance Python and C++ backend pipelines. Eager to bring robust prototyping, machine learning benchmarking, and Linux/CMake system development skills to the Valeo VITE R&D team.

CORE TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL, JavaScript
- **AI, GenAI & LLMs:** Google Gemini, OpenAI, LLAMA, RAG, Fine-Tuning, AI Agents, LangChain (LCEL), Prompt Engineering
- **Speech & Audio Tech:** Integration frameworks for Automatic Speech Recognition (ASR) & Text-to-Speech (TTS) pipelines
- **Machine Learning & Vision:** Supervised/Unsupervised Machine Learning, Deep Learning, Computer Vision, NLP, TensorFlow, PyTorch, OpenCV, YOLOv8, MediaPipe
- **Systems, APIs & Tools:** Linux (bash, ssh), FastAPI, Flask, REST APIs, CMake, Make, Git, GitHub, Docker, FAISS, OpenTelemetry, Arize Phoenix
- **Cloud & Platform:** Microsoft Azure, Cloud AI services
- **Automotive Protocols (Coursework/Familiarity):** SOME/IP, MQTT, protobuf, TCP/IP networking fundamentals

CERTIFICATIONS

- **Microsoft Certified:** Azure Fundamentals / AI Cloud Essentials —
https://drive.google.com/file/d/1UdwZK_rvRP2SYAc4CjQ1ug7d1e_5sGRV/view?usp=drive_link
- **Career Essentials in Generative AI:** Microsoft and LinkedIn —
https://drive.google.com/file/d/1eoWoMSQfeP7KU6_iD1jhcMzPvXlvCQf1/view?usp=drive_link
- **Microsoft Applied Skills:** Develop an agent with integrated tools (Microsoft)

PROFESSIONAL EXPERIENCE & R&D PROJECTS

NEXA RAG Platform Enterprise Document Intelligence —

https://github.com/Mohab3563/AI_Code_NEXA_RAG

Technologies: Python, FastAPI, Google Gemini, LLaMA, LangChain, FAISS, Arize Phoenix

- Designed, prototyped, and deployed an end-to-end RAG platform powered by Google Gemini and LLaMA, featuring an asynchronous FastAPI backend, CORS middleware, and dynamic persona

injection.

- Engineered a local embedding engine using HuggingFace all-MiniLM-L6-v2 on CPU with FAISS vector search, benchmarking model parameters for optimal retrieval performance.
- Implemented OpenTelemetry tracing and Arize Phoenix observability to profile, troubleshoot, and optimize application memory usage and P50/P99 latency.

FITAI AI Fitness & Nutrition Platform (Graduation Project - A+) —

https://www.linkedin.com/posts/mohab-rabie-246a1b36b_ai-machinelearning-yolo-activity-7340632587763142656-upEi

Technologies: Python, FastAPI, YOLOv8, MediaPipe, OpenCV

- Integrated three advanced AI pipelines: an AI conversational chatbot, real-time Exercise Form Analysis & Repetition Counter using MediaPipe, and Food Recognition & Calorie Estimation using YOLOv8.
- Developed scalable RESTful backend services using FastAPI for web and mobile clients, ensuring reliable, low-latency performance.

Youth Entities AI Chatbot - Engineers for Sustainable Egypt —

https://drive.google.com/file/d/12bV_ID2fEtEKgj-gUSqZQyzg2q50UvOo/view?usp=drive_link

Technologies: Python, FastAPI, Google Gemini, LLaMA, RAG, FAISS

- Built an intelligent context-aware chatbot providing real-time information processing and semantic retrieval by combining FAISS vector search with Google Gemini and LLaMA.
- Translated abstract user requirements into clear technical specifications, adhering to agile development and rapid feature iteration cycles.

TEACHING & CONTENT CREATION EXPERIENCE

AI & Programming Instructor | IBI Institute & 3C (Cairo, Egypt)

- Delivered bilingual instruction in Python, algorithms, and Machine Learning, effectively translating complex technical concepts into clear, structured specifications for students.

Online AI Instructor & Creator | Steam Park (Qatar - Remote) & YouTube —

<https://www.youtube.com/@HobaST/playlists>

- Conducted live virtual classes and published structured educational content covering Python programming and core Artificial Intelligence principles, cultivating strong analytical mindsets and communication skills.

EDUCATION

- **Master's Degree in Artificial Intelligence** | Cairo University (Cairo, Egypt)
- **Bachelor of Computers and Artificial Intelligence (Major: AI)** | Helwan University (Cairo, Egypt)
Status: 2025 Graduate | Grade: Excellent with Honors (Graduation Project Grade: A+)

LANGUAGES

- **Arabic:** Native
- **English:** Bilingual Proficiency (TOEFL Score: 500)
- **French:** Intermediate (B1)